

* Maxwell Eqs. in Media (macroscopically neutral)

$$\left\{ \begin{array}{l} \nabla \cdot \vec{D} = 0 \\ \nabla \cdot \vec{B} = 0 \\ \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \\ \nabla \times \vec{H} = \frac{\partial \vec{D}}{\partial t} \end{array} \right. , \quad \begin{array}{l} \vec{D} = \epsilon_0 \vec{E} + \vec{P} \quad (\vec{P}: \text{e dipole moment density} \\ \text{or polarization}) \\ \vec{B} = \mu_0 (\vec{H} + \vec{M}) \\ = \mu_0 \vec{H} \quad (\text{non magnetic media}) \end{array}$$

$$\nabla \times (\nabla \times \vec{E}) = -\frac{\partial}{\partial t} (\nabla \times \vec{B}) = -\mu_0 \frac{\partial}{\partial t} (\nabla \times \vec{H}) = -\mu_0 \frac{\partial^2 \vec{D}}{\partial t^2}$$

$$\nabla \times (\nabla \times \vec{E}) \equiv \nabla (\nabla \cdot \vec{E}) - \nabla^2 \vec{E} \quad , \quad \epsilon_0 \mu_0 = \frac{1}{c^2}$$

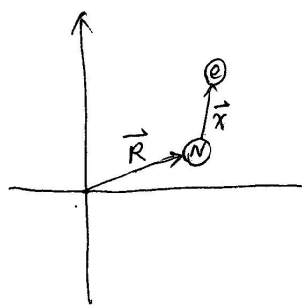
~~For transverse field~~ $\nabla \cdot \vec{E} = 0$

$$P \propto E$$

$$\therefore \boxed{\nabla^2 \vec{E} - \frac{1}{c^2} \frac{\partial^2 \vec{E}}{\partial t^2} = \frac{1}{\epsilon_0 c^2} \frac{\partial^2 \vec{P}}{\partial t^2}} \quad \text{eg (1)}$$

The specification of $\vec{P}$ depends on the property of the material or medium.

* Lorentz Model. (electron oscillator model)



$$|\vec{x}| \ll \lambda$$

$$m \frac{d^2 \vec{x}}{dt^2} = e \vec{E}(\vec{R}, t) + F_{en}(\vec{x}) \quad , \quad e \text{ is the charge of electron}$$

Assume the force between the electron and nucleus

$$F_{en}(\vec{x}) = -k_s \vec{x} \quad ,$$

$$\text{Define } \omega_0 = \sqrt{\frac{k_s}{m}} \quad , \quad \boxed{\left(\frac{d^2}{dt^2} + \omega_0^2 \right) \vec{x} = \frac{e}{m} \vec{E}(\vec{R}, t)} \quad (2.1)$$

The dipole moment of each electron $\vec{p} = e \vec{x}$.

The dipole moment density of the medium is $\vec{P} = N \vec{p}$, where N is the density of atoms

$$\therefore \boxed{\left(\frac{d^2}{dt^2} + \omega_0^2 \right) \vec{P} = \frac{Ne^2}{m} \vec{E}(\vec{R}, t)} \quad (2.2)$$

For linearly polarized wave, the electric field at the position of the atom is

$$\vec{E}(z, t) = \hat{e} E_0 \cos(\omega t - kz) \quad (3)$$

Assume E_0 is constant within the size of an atom when $|\vec{r}| \ll \lambda$

$$\nabla \cdot \vec{E} = 0.$$

For a single atom, from eq (2.1)

$$\left(\frac{d^2}{dt^2} + \omega_0^2\right) \vec{x} = \hat{e} \frac{e}{m} E_0 \cos(\omega t - kz) \quad (3.1)$$

$$\Rightarrow \boxed{\vec{x} = \hat{e} \left(\frac{e E_0 / m}{\omega_0^2 - \omega^2} \right) \cos(\omega t - kz)} \quad (4)$$

dipole moment: $\vec{p} = e \vec{x} = \alpha \vec{E}$, $\alpha(\omega) = \frac{e^2 / m}{\omega_0^2 - \omega^2}$: polarizability

$$\boxed{\vec{P} = N \vec{p} = N \alpha(\omega) \vec{E} = \hat{e} \left(\frac{N e^2 / m}{\omega_0^2 - \omega^2} \right) E_0 \cos(\omega t - kz)} \quad (5)$$

This is the polarization predicted by the Lorentz model when there is a field (3) in the medium. This solution should also satisfy Maxwell eq. (1)

eq (1) and (5) should be consistent.

$$\therefore \left(-k^2 + \frac{\omega^2}{c^2}\right) \hat{e} E_0 \cos(\omega t - kz) = -\frac{\omega^2}{c^2} \frac{N \alpha(\omega)}{E_0} \hat{e} E_0 \cos(\omega t - kz)$$

$$\boxed{k^2 = \frac{\omega^2}{c^2} \left(1 + \frac{N \alpha(\omega)}{E_0}\right) = \frac{\omega^2}{c^2} n^2(\omega)}$$

$$n^2(\omega) = 1 + \frac{N \alpha(\omega)}{E_0}, \quad \boxed{n(\omega) = \left(1 + \frac{N \alpha(\omega)}{E_0}\right)^{1/2}} \quad (6)$$

$$\boxed{V_p = \frac{\omega}{|k|} = \frac{c}{n(\omega)}}$$

this identifies $n(\omega)$ as the refractive index of the medium for light of frequency ω .

If we assume the Z electrons in an atom respond to the radiation independently.

$$\vec{x}_i = \hat{\epsilon} \left(\frac{e \epsilon_0 / m}{\omega_i^2 - \omega^2} \right) \cos(\omega t - kx), \quad \omega_i \text{ is the resonant frequency of the } i\text{th electron.}$$

the dipole mom. $\vec{p} = \sum_{i=1}^Z e \vec{x}_i$,

the polarizability $\alpha(\omega) = \sum_{i=1}^Z \frac{e^2 / m}{\omega_i^2 - \omega^2}$

polarization. $\vec{P} = \hat{\epsilon} N \sum_{i=1}^Z \left(\frac{e^2 / m}{\omega_i^2 - \omega^2} \right) \epsilon_0 \cos(\omega t - kx)$

refractive index. $n(\omega) = \left(1 + \frac{N \alpha(\omega)}{\epsilon_0} \right)^{1/2} = \left(1 + \frac{N}{\epsilon_0} \sum_{i=1}^Z \frac{e^2 / m}{\omega_i^2 - \omega^2} \right)^{1/2}$

For free electron, $\omega_0 = 0$

$$n(\omega) = \left(1 - \frac{N e^2}{\epsilon_0 m \omega^2} \right)^{1/2}$$

Define $\left(\frac{N e^2}{\epsilon_0 m} \right)^{1/2} = \omega_p$, plasma frequency.

When $\omega < \omega_p$, $n(\omega)$ is pure imaginary

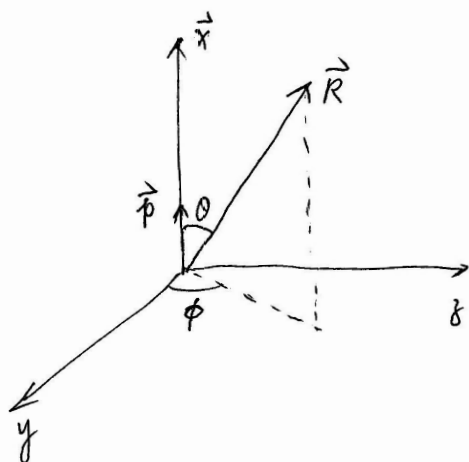
$$n(\omega) = \left(1 - \frac{\omega_p^2}{\omega^2} \right)^{1/2} = \frac{1}{\omega} (\omega^2 - \omega_p^2)^{1/2}$$

$$\mathcal{E}(z, t) = \hat{\epsilon} \epsilon_0 \cos \omega \left(t - n(\omega) \frac{z}{c} \right) = \text{Re} \left[\hat{\epsilon} \epsilon_0 e^{-i \omega \left(t - n(\omega) z / c \right)} \right]$$

$$(\omega^2 - \omega_p^2)^{1/2} = i(\omega_p^2 - \omega^2)^{1/2}, \quad \text{define } b(\omega) = \frac{(\omega_p^2 - \omega^2)^{1/2}}{c}$$

$$\mathcal{E}(z, t) = \hat{\epsilon} \epsilon_0 e^{-b(\omega) z} \cos \omega t, \quad \text{electric field decrease exponentially in the medium.}$$

E-Dipole radiation.



$$\vec{p} = \hat{z} p(t)$$

For $R \rightarrow \infty$.

$$\vec{E}(\vec{R}, t) = \frac{1}{4\pi\epsilon_0 c^2 R} \left[\underbrace{(\hat{z} \cdot \hat{R}) \hat{R}}_{\text{red circled}} - \hat{z} \right] \ddot{p}\left(t - \frac{R}{c}\right)$$

$$\vec{B}(\vec{R}, t) = \frac{1}{4\pi\epsilon_0 c^3 R} \left[\underbrace{\hat{z} \times \hat{R}}_{\text{red circled}} \right] \ddot{p}\left(t - \frac{R}{c}\right)$$

Poynting vector $\vec{S}$ gives the rate at which EM energy crosses
a unit area. From the dipole

$$\begin{aligned} \vec{S} &= \vec{E} \times \vec{H} = \epsilon_0 c^2 \vec{E} \times \vec{B} \\ &= \hat{R} \frac{\sin^2 \theta}{16\pi^2 \epsilon_0 c^3 R^2} \ddot{p}\left(t - \frac{R}{c}\right)^2 \end{aligned}$$

Total power flows out from the dipole at radius R is

$$\begin{aligned} P_{\text{wr}} &= R^2 \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin\theta \hat{R} \cdot \vec{S} \\ &= \frac{1}{4\pi\epsilon_0} \frac{2}{3c^3} \left[\ddot{p}\left(t - \frac{R}{c}\right) \right]^2 \end{aligned}$$

$$P_{\text{wr}} = \frac{1}{4\pi\epsilon_0} \frac{2}{3c^3} \frac{d^2 p}{dt^2}$$

In dilute gas

$$\vec{p} = \hat{x} \alpha(\omega) \epsilon_0 \cos \omega(t - \frac{z}{c}) \quad , \quad \text{assuming } n(\omega) \approx 1$$

$$p_{wr} = \frac{\omega^4}{6\pi\epsilon_0 c^3} \alpha^2(\omega) \epsilon_0^2 \cos^2 \omega(t - \frac{z}{c})$$

The income wave.

$$\vec{S} = \vec{E} \times \vec{H} = \epsilon_0 c^2 \vec{E} \times \vec{B}$$

$$\vec{E} = \hat{x} \epsilon_0 \cos \omega(t - \frac{z}{c})$$

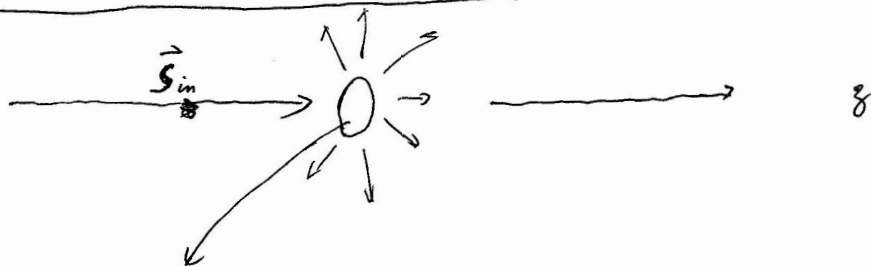
$$\vec{B} = +\frac{1}{c} \hat{y} \epsilon_0 \cos \omega(t - \frac{z}{c})$$

$$\vec{S}_{\text{incoming}} = \epsilon_0 c \hat{z} \epsilon_0^2 \cos^2 \omega(t - \frac{z}{c})$$

$$\therefore \epsilon_0^2 \cos^2 \omega(t - \frac{z}{c}) = \frac{1}{\epsilon_0 c} |\vec{S}_{\text{in}}|$$

$$p_{wr} = \frac{8\pi}{3} \omega^4 \left(\frac{\alpha(\omega)}{4\pi\epsilon_0 c^2} \right)^2 |\vec{S}_{\text{in}}|$$

Scattered power = cross section $\times$ incident power flux

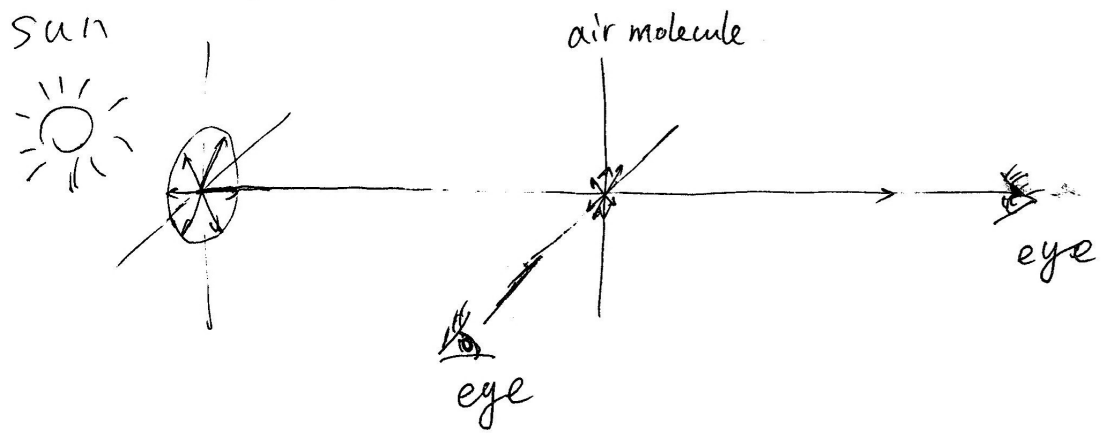


The area = scattering cross section.

$$\sigma(\omega) = \frac{8\pi}{3} \omega^4 \left(\frac{\alpha(\omega)}{4\pi\epsilon_0 c^2} \right)^2$$

$$\epsilon_0 [n^2(\omega) - 1] = N \alpha(\omega) \quad \left[\sigma(\omega) = \frac{8\pi}{3} \frac{\omega^4}{c^4} \left(\frac{n^2(\omega) - 1}{4\pi N} \right)^2 \right]$$

It is as if each scattering atom had a cross-sectional area σ , and all of the incident field power intercepted by this area were scattered.



Why sky light is blue and polarized.?

Absorption.

$$m \frac{d^2 \vec{x}}{dt^2} = e \vec{E}(\vec{R}, t) - m\omega_0^2 \vec{x} + \vec{F}_{fric}$$

$$\vec{F}_{fric} = -b \frac{d\vec{x}}{dt}$$

Define $\beta = \frac{b}{m}$.

$$\boxed{\frac{d^2 \vec{x}}{dt^2} + \beta \frac{d\vec{x}}{dt} + \omega_0^2 \vec{x} = \hat{z} \frac{e}{m} E_0 \cos(\omega t - kz)} \quad \text{eq. 18}$$

If a function and its derivatives are power 1 in the Diff. Egn.,

It's a linear D.E.

We can write $E_0 \cos(\omega t - kz)$ as $E_0 e^{-i(\omega t - kz)}$, and solve $\vec{x}(t)$.

$\vec{x}(t)$ will be complex. We take the real part of $\vec{x}(t)$ in the end.

$\text{Re}[\vec{x}(t)]$ is physically meaningful.

$$\left[\frac{d^2}{dt^2} + \beta \frac{d}{dt} + \omega_0^2 \right] \vec{x} = \hat{z} \frac{e}{m} E_0 e^{-i(\omega t - kz)} \quad (9)$$

try $\vec{x}(t) = \vec{a} e^{-i(\omega t - kz)}$

$$(-\omega^2 - i\beta\omega + \omega_0^2) \vec{a} = \hat{z} \frac{e}{m} E_0$$

$$\vec{a} = \frac{-\hat{z} (e/m) E_0}{\omega_0^2 - \omega^2 - i\beta\omega}$$

The physically meaningful solution is

~~(10)~~

$$\vec{x}(t) = \text{Re} \left(\frac{\hat{z} (e/m) E_0 e^{-i(\omega t - kz)}}{\omega_0^2 - \omega^2 - i\beta\omega} \right) \quad (10)$$

$$\omega_0^2 - \omega^2 = (\omega_0 + \omega)(\omega_0 - \omega) = 2\omega(\omega_0 - \omega) \quad , \quad \text{near resonant approximation.}$$

$$\vec{x}(t) = \hat{\epsilon} \left[U \cos(\omega t - k z) - V \sin(\omega t - k z) \right]$$

$$U = \frac{e E_0}{2m\omega} \frac{\omega_0 - \omega}{(\omega_0 - \omega)^2 + (\beta/2)^2}$$

$$V = \frac{e E_0}{2m\omega} \frac{-\beta/2}{(\omega_0 - \omega)^2 + (\beta/2)^2}$$

$$\text{Atom's power} = \vec{F} \cdot \dot{\vec{x}} = e \vec{E} \cdot \dot{\vec{x}} \quad , \quad \vec{E} = \hat{\epsilon} E_0 \cos(\omega t - k z)$$

$$\frac{dW_A}{dt} = e \omega E_0 \left[-\frac{1}{2} U \sin(2\omega t - 2k z) - V \cos^2(\omega t - k z) \right]$$

the ~~1st~~ 1st term ~~in~~ time averages to zero,

The 2nd term is always positive, (note e is negative here)

Atom's power is positive.

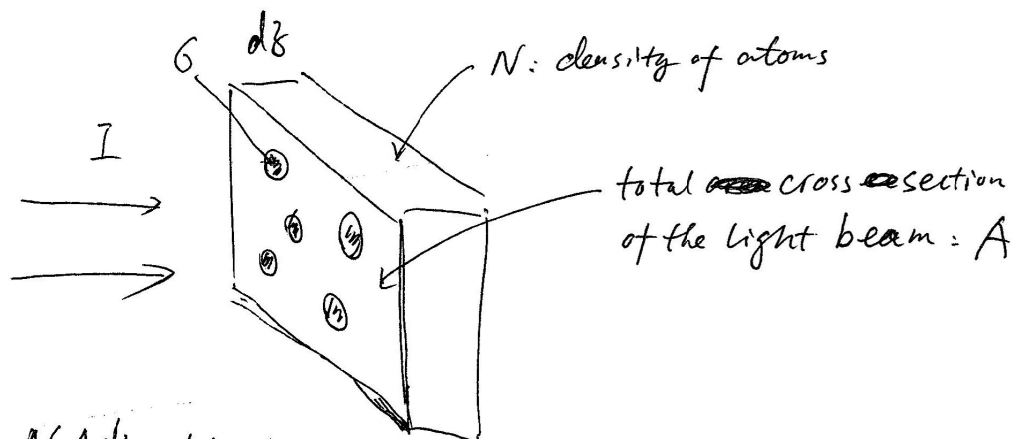
W_A : Atom's energy $\uparrow$. W_{em} : Field's energy $\downarrow$

$$\begin{aligned} \frac{dW_{em}}{dt} &= e \omega E_0 V \cos^2(\omega t - k z) \\ &= \frac{-e^2}{2m} \frac{\beta/2}{(\omega_0 - \omega)^2 + (\beta/2)^2} E_0^2 \cos^2(\omega t - k z) \\ &= \frac{-e^2}{2\epsilon_0 m c} \frac{\beta/2}{(\omega_0 - \omega)^2 + (\beta/2)^2} |\vec{S}_{in}| \end{aligned}$$

$$\vec{S}_{in} = \hat{\epsilon} \epsilon_0 c E_0^2 \cos^2(\omega t - k z)$$

We can also define an absorption cross section

$$\sigma(\omega) = \frac{e^2}{2\epsilon_0 m c} \frac{\beta/2}{(\omega_0 - \omega)^2 + (\beta/2)^2} \quad , \quad I_{\omega}(\beta) = I_{\omega}^{(0)} e^{-N \sigma(\omega) z}$$



$N \cdot A dz$ = total # of atoms in the volume.

$\sigma \cdot N A dz$ = total ~~area of~~ absorption cross section

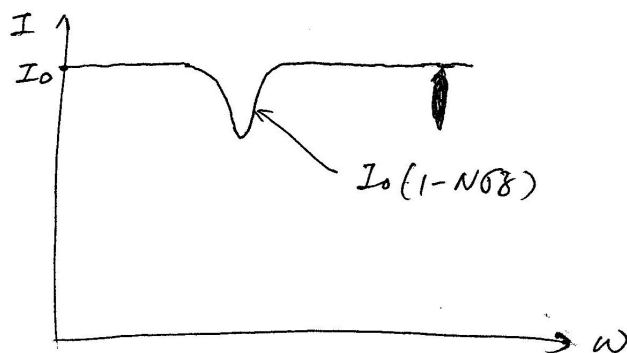
$$\frac{dI}{I} = -\frac{\sigma N A dz}{A} = -\sigma N dz$$

$$I = I_0 e^{-N\sigma z}$$

(Note: dz is so small that ~~no~~ atoms don't overlap ~~one~~ another)

~~For~~ where $N\sigma z \ll 1$

$$I = I_0 (1 - N\sigma z)$$



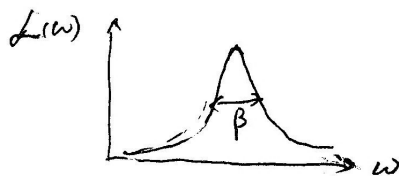
$$\sigma(\omega) = \frac{e^2}{2\epsilon_0 m c} \frac{\beta/2}{(\omega_0 - \omega)^2 + (\beta/2)^2}$$

Lorentz line shape.

$N\sigma(\omega)$

$$= \frac{Ne^2}{4\epsilon_0 m c} \cdot \frac{\beta}{(\omega_0 - \omega)^2 + (\beta/2)^2}$$

$$= \frac{Ne^2}{4\epsilon_0 m c} L(\omega)$$



$$\int_{-\infty}^{\infty} d\omega L(\omega) = 1 \quad \therefore \int_{-\infty}^{\infty} \sigma(\omega) d\omega = \frac{e^2}{4\epsilon_0 m c}$$

$$\sigma_{\text{t}} = \int_{-\infty}^{\infty} \sigma(\omega) d\omega = \frac{e^2}{4\epsilon_0 m c} = \pi r_e c$$

$$\sigma_{\text{t}}(\text{actual}) = \sigma_{\text{t}}(\text{classical}) \cdot f$$

f is called oscillator strength.

The Q.M origin of f ?

$$\vec{p} = e\vec{x}$$

$$= \alpha(\omega) \hat{\epsilon} \epsilon_0 e^{-i(\omega t - kx)}$$

$$\alpha(\omega) = \frac{e^2/m}{\omega_0^2 - \omega^2 - i\beta\omega}$$

$$\vec{p} = N\vec{p} = N\alpha(\omega) \hat{\epsilon} \epsilon_0 e^{-i(\omega t - kx)}$$

$$\nabla^2 \vec{\epsilon} - \frac{1}{c^2} \frac{\partial^2 \vec{\epsilon}}{\partial t^2} = \frac{1}{\epsilon_0 c^2} \frac{\partial^2 \vec{p}}{\partial t^2}$$

$$k^2 = \frac{\omega^2}{c^2} n^2(\omega)$$

$$n^2(\omega) = 1 + \frac{\alpha(\omega) N}{\epsilon_0}$$

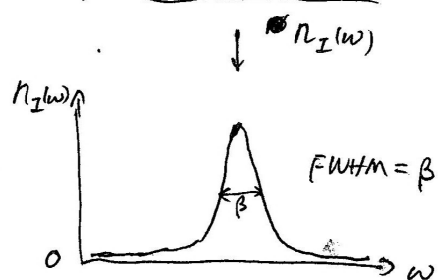
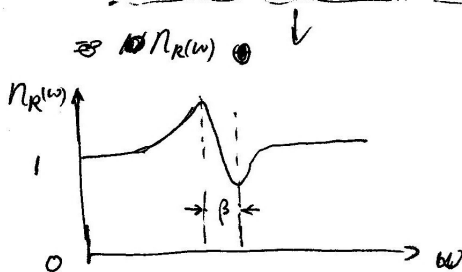
Use the approximation $|\omega_0 - \omega| \ll \omega, \omega_0$

$$\alpha(\omega) = \frac{e^2/2m\omega}{\omega_0 - \omega - i\beta/2} = \left[\frac{\omega_0 - \omega}{(\omega_0 - \omega)^2 + (\beta/2)^2} + i \frac{\beta/2}{(\omega_0 - \omega)^2 + (\beta/2)^2} \right] \frac{e^2}{2m\omega}$$

$$n(\omega) = \left(1 + \frac{\alpha(\omega) N}{\epsilon_0} \right)^{1/2}$$

$$\approx 1 + \frac{N}{2\epsilon_0} \alpha(\omega)$$

$$= 1 + \frac{Ne^2}{4\epsilon_0 m \omega} \frac{\omega_0 - \omega}{(\omega_0 - \omega)^2 + \beta^2} + i \frac{Ne^2}{4\epsilon_0 m \omega} \frac{\beta/2}{(\omega_0 - \omega)^2 + (\beta/2)^2}$$



The physical meaning of $n_I(\omega)$

$$\vec{E}(z, t) = \hat{e} \sum_{\omega} E_0 e^{-i(\omega t - k z)}$$

$$= \hat{e} \sum_{\omega} E_0 e^{-i\omega t - n_I(\omega) k_0 z/c}$$

$$= \hat{e} \sum_{\omega} E_0 e^{-n_I(\omega) \omega z/c} e^{-i\omega [t - n_R(\omega) z/c]}$$

$$= \hat{e} \sum_{\omega} E_0 e^{-n_I(\omega) k_0 z} e^{-i[\omega t - n_R(\omega) k_0 z]}$$

$$, k_0 = \frac{\omega}{c}$$

The Intensity $I_{\omega}(z)$

$$= \textcircled{9} \quad I_{\omega}(z) = I_{\omega}(0) \left[e^{-n_I(\omega) k_0 z} \right]^2$$

$$= I_{\omega}(0) e^{-2n_I(\omega) k_0 z}$$

$$2n_I(\omega) k_0 = \textcircled{9} \frac{Ne^2}{2\epsilon_0 mc} \frac{\beta/2}{(\omega_0 - \omega)^2 + (\beta/2)^2}$$

$$= N\sigma(\omega)$$